Internal Migration in the U.S. During the COVID-19 Pandemic

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Abstract

Survey and administrative internal migration data disagree on whether the COVID-19 pandemic increased or decreased mobility in the U.S. Moreover, though scholars have theorized and documented migration in response to environmental hazards and economic shocks, the novel conditions posed by a global pandemic make it difficult to hypothesize whether and how American migration might change as a result. We link individual-level data from the United States Postal Service's National Change of Address (NCOA) registry to American Community Survey (ACS) and Current Population Survey (CPS-ASEC) responses and other administrative records to document changes in the level, geography, and composition of migrant flows between 2019 and 2021. We find a 2% increase in address changes between 2019 and 2020, representing an additional 603,000 moves, driven primarily by young adults, earners at the extremes of the income distribution, and individuals (as opposed to families) moving over longer distances. Though the number of address changes returned to pre-pandemic levels in 2021, the pandemic-era geographic and compositional shifts in favor of longer distance moves away from the Pacific and Mid-Atlantic regions toward the South and in favor of younger, individual movers persisted. We also show that at least part of the disconnect between survey, media, and administrative/third-party migration data sources stems from the apparent misreporting of address changes on Census Bureau surveys. Among ACS and CPS-ASEC householders linked to NCOA data and filing a permanent change of address in their 1-year survey response reference period, only around 68% of ACS and 49% of CPS-ASEC householders also reported living in a different residence one year ago in their survey response.

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Introduction

Pandemic-era media narratives suggest that migration and residential mobility in the United States sky-rocketed in response to COVID-19. Though details vary across the nation, headlines commonly noted the unprecedented flow of people from large cities – and wealthier neighborhoods within those cities – to surrounding suburbs, exurbs, and beyond. Headlines from *The New York Times* are representative in this regard: "New Yorkers Are Fleeing to the Suburbs: 'The Demand is Insane'" (Haag 2020); "A Century-Old Moving Company Says the Summer of Covid is 'Insane'" (Rothman and Feinberg 2020); and "The Richest Neighborhoods Emptied Out Most as Coronavirus Hit New York City" (Quealy 2020). Pandemic-era data from multiple sources suggest that headlines like these, though sensationalized, are more than just hype. The "urban exodus" they pronounce – driven by increased financial instability and the desire to escape conditions ideal for COVID's spread (Cohn 2020, 2021), and fueled by a 28% drop in mortgage interest rates[1] and the newfound tolerance in some occupations for telecommuting (Hickman and Saad 2020) – increased demand for single-family housing across the country and drove home prices upward by 45% (Shiller 2022). Moreover, data from credit panels and administrative sources (discussed in detail below) confirm a shift in the intensity and direction of migration flows in favor of smaller, less dense, and more affordable places as the pandemic set in.

Contradicting this popular narrative, however, are estimates of migration and residential mobility from Census Bureau surveys which suggest that *fewer* Americans moved during the COVID-19 pandemic than before it. According to Current Population Survey Annual Social and Economic Supplement (CPS-ASEC) estimates, internal migration rates fell from 9.8% in 2019 to 9.3% in 2020 to 8.4% in 2021.[2] Though publicly available data from the American Community Survey are more limited, estimates from 2019 and 2021 also suggest an overall decline in migration from 13.7% to 12.8%, with intracounty mobility declining (from 7.5% to 6.7%) and trends in migration within and between states holding steady (approximately 3.3% and 2.4%, respectively).[3]

We examine migration patterns between 2019 and 2021 in hopes of better understanding both internal migration trends during the COVID-19 pandemic and the apparent disconnect between survey-derived estimates of accelerating migration decline on the one hand, and popular narratives backed by administrative data noting increased residential mobility during the pandemic on the other. Using linked administrative records from the United States Postal Service (USPS), Social Security Administration (SSA), and Internal Revenue Service (IRS), we compare

[1] Between January 2020 and 2021, the 30-year fixed rate mortgage average in the U.S. declined from 3.72% to 2.65%. Freddie Mac, 30-Year Fixed Rate Mortgage Average in the United States [MORTGAGE30US], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/MORTGAGE30US, June 3, 2024.
[2] Retrieved from https://www.census.gov/data/tables/time-series/demo/geographic-mobility/historic.html on November 18th, 2022.
[3] Retrieved from https://data.census.gov/table?q=International+and+Domestic+Migration&d=ACS+1-Year+Estimates+Subject+Tables&tid=ACSST1Y2019.S0701 and https://data.census.gov/table?q=International+and+Domestic+Migration&d=ACS+1-Year+Estimates+Subject+Tables&tid=ACSST1Y2021.S0701 on November 18, 2022.

pandemic-era address changes in 2020 and 2021 to pre-pandemic address changes in 2019. We argue that these linked migration data, rooted in the USPS's National Change of Address (NCOA) registry, hold untapped potential for understanding whether and how migration changed in response to COVID-19 because they (1) capture both *temporary* and *permanent* moves, (2) contain move dates which allow for analysis of within-year migration trends, and (3) allow the disaggregation of migration patterns by age, sex, race, ethnicity, and income when linked with other administrative and Census Bureau data. We use these novel linked data to answer some simple but fundamental questions. Did migration increase or decrease during the pandemic? When, where, for whom, and by how much? And, finally, why do migration trends in our linked administrative data (as well as in the popular narrative) contradict ACS- and CPS-ASEC-derived estimates?

Our linked data tell a mixed story about migration during the pandemic, but patterns are broadly more consistent with popular narratives and other research drawing on administrative and third-party data. We find a 2% increase in migration between 2019 and 2020 followed by a return to 2019 levels in 2021. Migration increased most in 2020 among individual movers, both the highest and lowest earners, those under age 25, and those moving longer distances across state or county lines. Family moves, moves among those 65 and older, and moves within counties declined, however. Though migration for most groups returned to 2019 levels in 2021, increased migration persisted through 2021 for younger movers and low earners making individual moves across state and county borders.

Though a full accounting of the gaps between survey- and administrative record-derived migration estimates is beyond our scope, we find that discrepancies represent, in part, the apparent misreporting of moves on Census Bureau surveys. We perform an additional linkage of ACS and CPS-ASEC migration responses directly to NCOA filings for those individuals who *both* responded to a survey *and* filed a change of address. Of all ACS householders filing a permanent change of address with USPS at some point in the 12-month ACS reference period, only 68% also report living in a different residence one year ago on the ACS. An even smaller share – 49% – of analogous CPS-ASEC householders making permanent address changes report a move on the March supplement. This misreporting – coupled with rising survey nonresponse, in general, and pandemic-specific nonresponse, in particular – appears to be the fundamental driver of the disconnect between survey, popular, and administrative accounts of American migration during the COVID-19 pandemic.

Theoretical Framework and Prior Research

Vulnerability, Disasters, and Migration

Adjudicating opposing accounts of pandemic-era migration is complicated by the fact that extant migration theory and empirical research offer limited insight into how a global pandemic might influence individual migration propensities and aggregate migration patterns in the United States. Migration decisions are complicated even before pandemic-related concerns are factored in.

Households and individuals considering a move weigh, idiosyncratically, the economic, social, and psychological costs and benefits of migration and ultimately move if that complicated calculus works out in their favor (Greenwood 1985, 1997; Lee 1966; Ritchey 1976). But the pandemic introduces several unknowns into this complicated migration equation, making it difficult for movers to assess risk factors at both the origin and destination and know whether a move would benefit them or not. Moreover, the migration-relevant information available to potential movers changed rapidly and unpredictably as the pandemic unfolded. How should the risk of COVID-19 infection factor into the decision-making process? Will employers' newfound tolerance for telework survive the pandemic? How long will mortgage interest rates stay low, and how high will home prices rise? And so on. The shifting context in which individuals made migration decisions during the pandemic makes predicting aggregate changes in the level, composition, and direction of migration flows difficult.

Research on migration in response to natural disasters may offer some insight into migration during the COVID-19 pandemic. A growing literature documents the mobility of Americans from areas affected by severe hurricanes and other life threatening environmental hazards (e.g., DeWaard, Johnson, and Whitaker 2020; Fussell et al. 2017; Hunter 2005; Hunter et al. 2003; Morrow-Jones and Morrow-Jones 1991; Riosmena, Nawrotzki, and Hunter 2018), the intensity and frequency of which are increasingly linked to human activity and climate change (IPCC 2023; McGuire, Mason, and Kilburn 2002; Stott 2016). Broadly, these studies show that natural disasters displace affected residents, forcing them to nearby destinations, though often only temporarily. Many displaced residents return once disaster damage has been repaired (Curtis et al. 2020; DeWaard et al. 2020; Fussell, Sastry, and VanLandingham 2010), often alongside new migrants attracted by the economic opportunities introduced by the "recovery machine" in disaster-affected places (Pais and Elliott 2008). Migration following natural disasters tends to intensify and reinforce established migration flows (Fussell et al. 2017), lending credence to the notion that, although migration often appears volatile relative to other demographic processes, "once an overall size of migration is controlled for, spatial distribution patterns are found to be remarkably stable (Willekens and Baydar 1983:4)."[4]

Despite the surprising stability and predictability of macro-level population movements following chaotic natural disasters, research finds significant demographic and socioeconomic selection in terms of both flight and return migration. Broadly speaking, these patterns highlight the role of vulnerability in individual and household migration decisions, and they make explicit subgroup variation in vulnerability often hidden at the population level. Migration from natural disasters, though often mandated at the state or federal level, increases with (perceived) risk to person and property but may be mediated by the resources available to potential migrants (Hunter 2005). Women, children, elderly populations, ethnoracial minorities, and the socioeconomically disadvantaged are often most vulnerable yet least equipped to flee (Hunter 2005; Morrow-Jones and Morrow-Jones 1991). Despite the dearth of resources for making a move, research typically finds that, in the absence of an evacuation mandate, migrants fleeing

[4] See also Mueser (1989: 186) and Herting, Grusky, and Van Rompaey (1997: 268), respectively, on "regularities in the streams of migration" and the "intrinsic structuration" of established migration networks.

environmental hazards are negatively selected with respect to socioeconomic characteristics (Morrow-Jones and Morrow-Jones 1991). Research on return migration following disasters, however, typically finds that migrants are *positively* selected with respect to socioeconomic status. For example, migrants returning to areas affected by Hurricane Katrina were more likely to be white, college educated, and higher income; this positive selection, in turn, is largely explained by the fact that racial and ethnic minority, less educated, and lower income evacuees were more likely to live in areas prone to flooding and the complete destruction of homes (Fussell et al. 2010). As such, vulnerability and socioeconomic status shape aggregate patterns of both flight from and return migration to areas impacted by environmental hazards.

In addition to threatening physical security, the pandemic also resulted in economic and employment shocks more geographically dispersed than excess mortality at the start of the pandemic (Polyakova et al. 2020) and more akin, perhaps, to the Great Recession. Generally speaking, economic recessions dampen internal migration (Greenwood, Hunt, and McDowell 1986; Molloy, Smith, and Wozniak 2011). During the housing market collapse and associated economic recession of the late 2000s, migration over longer distances plummeted while local mobility increased (Stoll 2013). Though migration between labor markets is typically one way of dealing with unemployment, the recession impacted the entire nation and left those newly unemployed with no viable alternatives elsewhere and no economic incentives to migrate (Frey 2009; Stoll 2013). Moreover, the housing crisis and bursting of the housing bubble left many homeowners underwater, tied financially to homes worth less on the market than the current mortgage balance (Immergluck 2009). Echoing findings from the environmental hazard migration literature, Stoll (2013) notes that those most likely to move during the Great Recession lull were precisely those rendered most vulnerable by it – those recently unemployed and/or facing foreclosure and those targeted by subprime lenders (see Rugh and Massey 2010) – as they searched locally for affordable housing.

What disaster migration could tell us about migration during the pandemic

The literature describing the effects of economic and environmental disasters on migration, summarized in Table 1, suggests that the COVID-19 pandemic may influence (1) the overall level of migration, (2) the composition of the migrant population, and/or (3) the geographic distribution of migrants – largely through socioeconomic status and the ways in which some population subgroups are rendered more vulnerable by it than others. That said, it is not clear that the pandemic is directly comparable to environmental disasters or other recent economic downturns and, therefore, whether pandemic-era migration will have similar signatures to these events.

To the extent that the pandemic threatens potential movers' physical security in the ways natural disasters do, we might expect an overall increase in migration fueled by pandemic flight from areas hardest hit by COVID-19 to nearby destinations typically chosen by outmigrants, particularly among those at highest risk of exposure to and/or complications from COVID-19 (Razzaghi et al. 2020; Rothwell and Smith 2021; Selden and Berdahl 2020; Shortreed et al.

2023; Wiemers et al. 2020). This vulnerability effect may be tempered to the extent that those most vulnerable lack the socioeconomic resources to make an unexpected move. However, although they both pose existential threats to physical security, natural disasters and pandemics may not be directly comparable with respect to their likely effects migration. Natural disasters and emergency mandates are typically localized and do not affect all Americans simultaneously, giving those fleeing options for relocation nearby. Disasters also entail the physical destruction of homes and amenities which threatens security in much more direct ways than the lockdowns, mask mandates, store closures, and supply chain disruptions precipitated by COVID-19 – factors which, though inconvenient and stressful, nonetheless leave physical structures intact. Given these differences, it is plausible that the pandemic may have induced vulnerable Americans to forego migration and stay put even in areas hit hardest by COVID-19, potentially even resulting in an inversion of the positive relationship between vulnerability and migration typically witnessed in the environmental hazards literature.

Table 1: Summary of the Disaster Migration Literature and Pandemic Migration Hypotheses

	Environmental Hazards Migration	**Great Recession Migration**	**Pandemic Migration**
Aggregate Level of Migration	Increased migration flows, especially in local areas directly impacted	Reduced migration	*?*
Geography of Migration Flows	Migration away from hazards reinforces existing geographic patterns	Shift away from long-distance migration toward local residential mobility	*Migration from denser urban areas conducive to spread of COVID-19 to nearby areas*
Migrant Selection, Composition of Migrant Flows	Function of vulnerability and resources: *Flight* – migrants tend to be negatively selected; *Return* – migrants tend to be positively selected.	Increased mobility among those foreclosed upon and/or recently unemployed	*Increased mobility among those of means, those suffering economically, and/or those at greatest risk from COVID-19*

Similarly, to the extent that the economic effects of the pandemic were felt across the entire nation, we might expect a sharp decline in migration like that seen in the recent Great Recession, with the potential for local or mid-distance mobility among those most heavily impacted by the economic downturn. But, again, the economic insecurity precipitated by the pandemic differs in important ways from that of the Great Recession. Although unemployment increased in both cases, the pandemic saw workers in some occupations gain increased flexibility and

opportunities for telecommuting (Dingel and Neiman 2020). Additionally, rather than falling as they did in the late 2000s, housing prices increased during the pandemic, pandemic-era moratoria on evictions (at least temporarily) made housing more secure for those in the most precarious situations, and Economic Impact Payments and more generous unemployment benefits blunted the impacts of the economic downturn for many vulnerable Americans. As such, the increased flexibility gained by relatively advantaged workers and the less tenuous economic situation of the relatively disadvantaged may reverse the socioeconomic gradient on migration propensities witnessed during the Great Recession.

Prior Research on Pandemic-Era Migration

As noted above, Census Bureau survey-derived estimates of migration suggest that Americans approached the existential physical and financial threats posed by the pandemic by hunkering down. Faced with new unknowns, potential movers opted to stay put, moving less often than they did in the recent past. As Frey (2021a) notes in his analysis of CPS-ASEC data, the pandemic-era migration decline – driven primarily by a decline in family- and employment-related moves and by declining migration among millennials saddled with housing crises and underemployment at key moments in the life cycle – merely caps off a long-term "great American migration slowdown" dating back to the mid-20th century (Frey 2009).

But studies to date relying on administrative, commercial, and other data confirm popular migration narratives and suggest that Americans – or at least segments of the U.S. population – moved at *higher* rates during the pandemic than in the years preceding it. An analysis of cell phone geolocation and Facebook friendship data by Coven, Gupta, and Yao (2020), for example, finds evidence of large-scale flight from dense, urban areas like New York City during the onset of the pandemic, particularly among younger, wealthier, and white populations. Similarly, an analysis of USPS National Change of Address (NCOA) data finds high levels of out-migration from the wealthiest quintile of NYC tracts, as well as from tracts with large young adult populations (Quealy 2020). Pew Research Center surveys in June and November of 2020 find a similar age gradient on COVID-19 driven migration patterns, but suggest that white individuals were less likely than their Black, Hispanic, and Asian counterparts to have made COVID-19-related moves (Cohn 2020, 2021). Moreover, these surveys suggest that COVID-19-related financial pressures increasingly precipitated migration as the pandemic wore on (ibid.), supporting the notion that those with the means to escape dense environments conducive to COVID-19's spread in early 2020 did so.

Though particularly prominent in the nation's largest cities, apparent flight from larger, denser, and more expensive urban places proved ubiquitous. In an analysis of Census Bureau intercensal population estimates, Frey (2021b) notes that the pandemic exacerbated a longer-term geographic shift in net migration from major metropolitan areas to smaller metros and non-metros, as well as from urban core counties to suburban counties, across the U.S. Brown and Tousey (2021) find similar trends in their analysis of the Federal Reserve Bank of New York/Equifax Consumer Credit Panel (CCP) migration data, noting an increase in the volume of

migration between urban centers and suggesting that the patterns appear to be driven by the rapidly rising cost of living in larger urban areas. Examining data from Zillow and the USPS, Ramani and Bloom (2021) describe a similar flight of populations from dense urban centers to suburbs and exurbs, dubbing it the "donut effect" – a hollowing out of the urban core in favor of the peripheral ring around it. More recent analysis of Census Bureau intercensal population estimates suggests these trends persisted through 2022 (Rogers, Perry, and Spell 2023).

Decomposing CCP net migration into in- and out-flows, however, Whitaker (2021a, 2021b) finds that declines in net migration in urban neighborhoods were achieved primarily via declines in in-migration that outpaced increases in out-migration. As such, characterizing declining net migration in urban areas as flight may miss the mark: though out-migration from urban places increased in 2020, an even greater number of people who would have moved into an urban area in the counterfactual absence of the pandemic either did not move or moved to smaller urban, suburban, and/or exurban areas instead. An analysis of USPS NCOA data by MyMove.com suggests that the latter scenario is more likely, finding that the number of address changes between February and July of 2020 increased by nearly 4% over the same period in 2019 (Bowman 2020). Taken together, these alternative sources of migration data suggest that – rather than hunkering down – an increasing share of Americans continued to move during the pandemic due to COVID-19 risk and economic instability, but shifted migration inflows away from major metros and toward smaller destinations across the country.

As such, while Census Bureau surveys suggest that internal migration slowed in response to the existential and financial uncertainty introduced by the pandemic, other sources support the popular media narrative that migration increased in ways not necessarily anticipated by prior research on migration during economic recessions and environmental disasters. Knowing whether and how migration changed in response to the COVID-19 pandemic is important not only for our broader understanding of migration's determinants and effects, but also for our assessment of the efficacy of survey migration items in capturing migration behavior accurately. The analyses presented here not only bring another source of data to bear on the question of pandemic-era migration in the U.S., but also offer an assessment of the degree to which ACS and CPS-ASEC responses capturing migration during the pandemic agree with other administrative data capturing address changes over the same period.

Data

Measuring Migration in the NCOA Registry

Our primary source of migration data for the 2019 to 2021 period is the United States Postal Service's (USPS) National Change of Address (NCOA) registry. This registry includes all voluntary submissions made by individuals, families, and businesses in need of temporary or permanent mail forwarding services offered by the USPS.[5] The data contain the details of mail

[5] Mail forwarding requests can be made either in person at a post office or online at https://moversguide.usps.com/mgo/disclaimer?referral=UMOVE.

forwarding requests, including name, origin and destination address, type of address change (e.g., temporary/permanent and individual/family/business moves), and the date mail forwarding services should begin.[6] Updates to the registry are delivered to the Census Bureau monthly, and we use registry updates from October 2018 through December 2022 to understand migration patterns between January 2019 and December 2021.[7] Analysis throughout is restricted to unique[8] individual and family moves whose origin and destination are in the 50 states and Washington, D.C.

The NCOA data are particularly well-suited to understanding migration during the pandemic because they include precise move dates and both temporary and permanent address change requests filed with the USPS. Move dates allow us to see whether and how migration changed not only between 2019 and 2021, but also how it changed from month to month as the pandemic began in early 2020 and unfolded thereafter. Temporary moves (i.e., address changes that are likely to last fewer than 6 months and that are expected to end in a return to the original address) captured in the NCOA data have the unique potential to offer insights into the migration behavior of Americans weighing new, varied, and changing migration-relevant information as the pandemic set in.

A priori, the relationship between temporary and permanent migration is unclear. On the one hand, the two types of moves could be largely unrelated; temporary migration could capture moves among seasonal workers or those splitting their time between multiple homes, while permanent moves capture something more akin to the survey migration concept of one's "place of usual residence". In such a scenario, temporary and permanent migration would change largely independently of one another: one could increase while the other decreases, or both could increase or decrease in tandem, but for different reasons. On the other hand, particularly in a pandemic or disaster context, temporary and permanent migration could be interchangeable; for example, temporary migration could serve as a precursor to permanent migration, in which case the former could increase at the expense of the latter. We take an agnostic stance on the relationship between temporary migration and present all statistics separately for temporary and

[6] Note that the date mail forwarding services begin does not necessarily correspond with the date a change of address is filed. Movers can post-date a change of address filed after they move, but they can also file a change of address prior to moving with a request for forwarding services to begin in the future. For this reason, we date moves using the date forwarding services are requested to begin and (lacking any method for testing this assumption) assume throughout that this date more accurately captures when a given move occurred than the date a change of address was filed.

[7] Analysis is restricted to the 2019 to 2021 period for two reasons. First, the format of NCOA data deliveries changes substantially in October 2018, making the consistent cleaning and deduplication of the registry difficult to maintain prior to that delivery. Second, there is evidence of a lag in change of address requests. While some requests are filed prior to a move, others are filed after a move occurs. So, we use registry updates from October 2018 through December 2022 to get a comprehensive look at address changes made between January 1, 2019 and December 31, 2021.

[8] Movers can (and often do) file multiple changes of address for a single move. Additionally, USPS processing of address changes appears to generate duplicate rows in some cases. We deduplicate the NCOA data in an attempt to analyze unique moves by dropping rows flagged by USPS as "change" or "delete" transactions (as opposed to "adds" – new address change requests) and any remaining rows which duplicate the earliest observation for a given PIK, move date, origin ZIP code, and destination ZIP code combination.

permanent address changes. We note, though, that if temporary address changes were used as a substitute for permanent relocation early in the pandemic, temporary moves captured in the NCOA, but not survey, data may help explain the apparent disconnect between permanent changes in place of residence measured in Census Bureau surveys and the media narratives surrounding pandemic migration.

Despite this potential, and though the NCOA data are increasingly used among migration scholars in concert with other data, we know of no peer-reviewed studies in which they are used as a stand-alone source of migration data. The hesitancy to use the NCOA data as a stand-alone migration data source is, we suspect, attributable to the difficulties inherent in working with the data, the voluntary nature of the NCOA registry, and the potential for biases associated with self-selection in the NCOA universe (Plyer, Bonaguro, and Hodges 2010). We are not aware of any analyses quantifying this selection bias, much less of any changes in this bias over time. We also forego assessing selection into the NCOA universe here because doing so is complicated by substantial and contemporaneous increases in non-response in the very surveys sources one might use as a point of comparison (Rothbaum et al. 2021; Rothbaum and Bee 2021). Were we to conduct a selection analysis using these data, it would be impossible to distinguish a true change in NCOA selection bias from a change in the survey and migration item response rates of (non-)movers. Lacking a viable point of comparison, then, we acknowledge that there are biases present in the NCOA data – that is, not all movers file a change of address – but we assume throughout that these biases are consistent over time – that is, contingent on migrating, the factors determining selection into the NCOA data prior to the pandemic were the same as those during it. This assumption, while not ideal, is supported by analyses from Ramani and Bloom (2021) which demonstrate a consistently positive correlation between county-level NCOA migration flows and flows from 5-year ACS responses, records from Data Axle, and Census Bureau estimates constructed from IRS tax return data.

Measuring Age, Race, Ethnicity, and Income

We supplement NCOA migration data with age from the Social Security Administration's (SSA) Numerical Identification Database (Numident) and tax year 2020 (calendar year 2021) Adjusted Gross Income (AGI) from Internal Revenue Service (IRS) Forms 1040. The Numident file contains demographic information on all individuals ever issued a Social Security Number, and we construct age from Numident date of birth. Tax year 2020 AGI from IRS 1040s is used to rank individuals into deciles relative to their peers born in the same year.[9]

Race and Hispanic origin are drawn from a composite file built by Census Bureau researchers from a broad array of decennial, survey, and administrative record data (Ennis et al. 2018). We use this file to sort individuals into one of nine mutually exclusive ethnoracial categories: Hispanic (any race), non-Hispanic (NH) White Alone, NH Black or African American Alone

[9] We use tax year 2020 (filing year 2021) 1040s because Coronavirus Aid, Relief, and Economic Security (CARES) Act tax credits increased filing incentives and, therefore, potentially the coverage of the U.S. population relative to other years. Non-filers are not ranked but retained in their own separate category.

(Black hereafter), NH American Indian or Alaskan Native Alone (AIAN), NH Asian Alone, NH Native Hawaiian or Other Pacific Islander Alone (NHPI), NH Some Other Race (SOR) Alone, NH Multiple Race, and Missing race and ethnicity.[10]

ACS and CPS-ASEC Migration Responses

Though the bulk of our analysis utilizes the linked NCOA, SSA, and IRS administrative data described above, we also present results in which individuals in the NCOA registry are linked to their ACS and CPS-ASEC survey responses. Both the ACS and CPS-ASEC provide valuable migration information derived from responses to "residence one year ago" questions, but they differ from one another and from NCOA data in several ways relevant to our analysis.[11] First, the surveys differ with respect to the precise measurement of "one year ago". While the CPS-ASEC checks consistently for a move since March of the previous year, the ACS features a rolling survey window which shifts depending on the month in which each individual responds. For each survey, we restrict all NCOA filings for a given individual to those made in the year preceding survey response, but these time periods differ for ACS respondents, while they are strictly March-to-March for CPS-ASEC respondents. Second, the surveys define residency differently. While the ACS uses a "two month" rule to establish residency – thereby potentially counting as migrants those moving temporarily but for longer than two months – the CPS-ASEC asks respondents to consider their "place of usual residence." Given these differences in residency definitions, we restrict this portion of our analysis to *permanent* changes of address so that survey and change of address migration align as closely as possible conceptually. Third, though response rates in both surveys have been falling and both surveys experienced pandemic-related data collection issues, both unit and migration item non-response are higher in the CPS-ASEC than in the ACS. As a result, CPS-ASEC relies more heavily on allocation and imputation procedures to fill in missing migration responses. We restrict our analysis to migration item respondents to avoid any contamination of our results by allocation and imputation errors. Finally, note that name limitations on the change of address data require further restrictions to our individual NCOA-survey migration comparisons. USPS only requests a single name for families changing addresses together: a single change of address form covers everyone receiving mail at the address with the same surname as the filer. Given this limitation, we restrict this portion of our analysis to survey householders only.[12]

[10] Prior research from Ennis and colleagues (2018: 187) suggests that those missing a race and ethnicity in this composite dataset are more likely to be racial and ethnic minorities, male, under age 18, renters, members of larger households, and/or residents of tracts with higher median incomes.

[11] See Ihrke, Koerber, and Fields (2015) for a detailed accounting of migration-relevant differences between the ACS and CPS-ASEC.

[12] Implicit in this restriction is an assumption that survey householders are those most likely to file a change of address, and vice versa. In supplemental analyses not reported here, we relaxed this restriction and compared the migration responses of all individuals found in both the NCOA registry and either ACS or CPS-ASEC. These supplemental analyses found much lower rates of survey migration reporting among NCOA movers than reported in the main text below, suggesting that survey householders are less likely to report permanent address changes made by others in the household if they, themselves, did not move.

The goal of this additional linkage and analysis is to understand whether and how discrepancies between administrative record and survey accounts of migration may be, at least in part, the result of discrepancies in migration reporting *at the individual level*. Research on this possibility is scant, but prior studies provide an indication that changes of address found in administrative records may not be accurately captured in survey responses. First, Hyatt and colleagues (2018: 2177) linked CPS-ASEC migration responses with Longitudinal Employer-Household Dynamics (LEHD) address data and found that nearly half "of CPS respondents who are cross-state migrants in the administrative record data do not report a cross-state move in the CPS and that this disagreement has grown over time." Second, Foster, Ellis, and Fiorio (2018a, 2018b) linked migration measures derived from IRS 1040s and information returns (1099s, W-2s, etc.) to ACS responses and found that 14% of all ACS respondents reporting no move in the last year changed the state from which they filed taxes (and/or received information returns) over the same period. Our work builds upon these studies and brings more nuanced change of address data, rather than data collected for tax purposes, to bear on the accuracy of survey migration measures.

Data Linkage

Migration data from NCOA are linked with the Numident, IRS 1040s, and Census Bureau surveys using unique, anonymous identifiers assigned to individuals in each dataset. These Protected Identification Keys (PIKs) are assigned by Census Bureau using Person Identification Validation System (PVS) software (Wagner and Layne 2014). PVS software uses personally identifiable information (PII) such as SSN and name, as well as date of birth and address, to assign individuals a PIK. Once PIKs are assigned and before researchers gain access to the files, all PII is scrubbed from the data to maintain confidentiality.

Not all individuals found in administrative and survey records can be successfully assigned a PIK. PIK assignment rates are near 100% in data sources containing SSNs (e.g., the Numident and IRS 1040s) but are lower in data sources without them (e.g., surveys and the NCOA registry). Prior analysis shows that successful PIK assignment in data sources without SSNs is non-random: young children, racial and ethnic minorities, immigrants, low-income individuals, non-employed individuals, and – key for this analysis – recent movers are less likely to receive a PIK than other individuals (Bond et al. 2014). Crucially, those not assigned a PIK cannot be linked to other data sources.

Incomplete PIK assignment is problematic not only because it introduces additional bias into the universe of individuals self-selecting into the NCOA registry, but also because NCOA PIK assignment rates increase from 66% in 2019 to 69% in 2020 to 75% in 2021 (Table 2). Failure to correct for this rising PIK rate would, regardless of trends in all NCOA filings, produce a steadily rising migration trend among the subset of linkable records. Moreover, the lower probability of PIK assignment among movers inhibits our ability to observe the population we are interested in.

To account for incomplete and variable PIK assignment rates in the NCOA data, we create weights for individuals successfully assigned a PIK as follows. First, we place all change of address observations in mutually exclusive bins delineated by move details captured on NCOA records, including move month and year, origin and destination county, temporary vs. permanent address change, and individual vs. family move. Second, we calculate the PIK rate within each month/year, origin/destination, temporary/permanent, and individual/family-specific bin. Finally, we weight PIKed observations in each bin by (bin-specific PIK rate)$^{-1}$, such that the sum of all weighted PIKed observations in each bin equals the sum of all unweighted PIKed and unPIKed observations in each bin.

Note that while these weights address bias in PIK assignment among those in the NCOA data with respect to the move characteristics captured in the data, they do not *explicitly* address other known biases in PIK assignment by demographic and socioeconomic characteristics (Bond et al. 2014). To the extent that characteristics available on the NCOA data (such as origin and destination county) are correlated with these other characteristics (via racial, ethnic, and/or economic segregation, for example), our weighting scheme may *implicitly* correct for these biases. Regardless, results presented below relying on links between NCOA and other data sources should be interpreted with caution, as they may underestimate moves among groups less likely to be assigned a PIK.

Methods

Internal Migration during the COVID-19 Pandemic

Did migration increase or decrease during the pandemic? When, and by how much? Does the direction and magnitude of pandemic-era migration changes differ by the characteristics of the move and/or movers and, if so, how?

Answering these questions in an absolute, direct way is complicated by the fact that not all movers file a change of address with the USPS. However, as discussed above, we operate under the assumption that the factors influencing self-selection into the NCOA data are consistent over time and look at year-over-year changes in NCOA filings. In other words, we compare address change counts from 2020 and 2021 to address changes in 2019. Comparisons for a given sociodemographic group or type of move, *g*, are presented throughout as ratios:

$$\frac{2020AddressChanges_g}{2019AddressChanges_g} \text{ and } \frac{2021AddressChanges_g}{2019AddressChanges_g},$$

where a ratio of 1.0 indicates no change in migration relative to 2019, a ratio greater than 1.0 indicates an increase in migration, and a ratio less than 1.0 indicates a migration decline. We present ratios for the nation, as well as by month, age, race and ethnicity, within-birth-cohort AGI rank, geography, type of move (temporary vs. permanent, individual vs. family), and move scale (intracounty vs. intercounty/intrastate vs. interstate).

Comparing Individual Changes of Address and Survey Migration Responses

At the individual level, does migration as captured by permanent change of address filings agree with responses to migration items in the ACS and CPS-ASEC?

As noted above, we answer this question by linking individuals who filed a permanent change of address to their ACS or CPS-ASEC responses from 2020 and 2021. We restrict analysis to those linked householders[13] whose permanent change of address occurs in the 1-year migration interval referenced in their ACS or CPS-ASEC response. We report the share of all householders making permanent changes of address who also accurately report that permanent move on their survey response.

Results

Aggregate Change of Address Trends, 2019-2021

Rather than a sustained upward or downward trend in migration, neatly adjudicating the survey-media disagreement framing our analysis, the National Change of Address (NCOA) data tell a mixed story. As shown in Table 2, and consistent with popular narratives and non-survey data sources, address changes increased by 2% from 30.6 to 31.2 million between 2019 and 2020. However, consistent with survey trends, address changes subsequently fell below 2019 levels to 30.3 million in 2021 (a 1.1% decrease relative to 2019). Broadly speaking, these aggregate patterns suggest that Americans responded to the novel threats posed by COVID-19 in 2020 by moving more often – akin to migration following a natural disaster – but moved less often in 2021 as the economic effects of the pandemic spread.

It is important to note that the 2% increase in changes of address we find in 2020 is not driven primarily by moves unlikely to be captured in Census Bureau surveys, such as temporary or repeat address changes. Although percentage increases in 2020 are larger among temporary (6%) than among permanent (2%) moves, increased temporary migration accounted for only 28.7% of the overall increase in address changes (Table 2). Similarly, although the share of all address changes made by repeat movers increased from 7.6% to 8.1% between 2019 and 2020, this increase accounts for only 38% of the overall increase in address changes over the same period; the majority of the 2020 migration spike is attributable to individuals moving for the first time that year. The NCOA data show evidence of increases in migration even among first-time permanent movers in the pandemic's first year that simply do not seem to be captured in the ACS or CPS-ASEC.

[13] For family movers, change of address forms require a single name and all mail addressed to anyone in the household with the same last name will be forwarded. We assume, for this analysis, that the survey householder and the person listed in the change of address form are one and the same.

Table 2: Temporary and Permanent Address Changes, 2019 to 2021

	2019	2020	2021
NCOA PIK Rate	66.3%	69.2%	75.2%
Total Address Changes	30,635,000	31,238,000	30,300,000
Temporary	2,955,000	3,128,000	2,810,000
Permanent	27,680,000	28,110,000	27,490,000
Total Address Changes relative to 2019	--	1.02	0.99
Temporary	--	1.06	0.95
Permanent	--	1.02	0.99
Repeat Mover Share	7.6%	8.1%	8.6%
Year-Over-Year Change	--	603,000	–938,000
Year-Over-Year Change Attributable to Repeat Movers	--	38.4%	–7.3%

Source: National Change of Address registry (2019-2021). CBDRB-FY2022-CES005-018.
Notes: Counts of PIKed NCOA observations, weighted such that their sum is equal to the count of all PIKed and unPIKed observations, are shown. This weighting scheme corrects for changes in PIK rates over time. Per Census Bureau Disclosure Review Board requirements, temporary and permanent mover counts are rounded to no more than four significant digits.

Table 3: Intracounty, Intercounty, and Interstate NCOA Address Changes, 2020 and 2021 relative to 2019

	Temporary		Permanent		Overall	
	2020	2021	2020	2021	2020	2021
Nationally	1.06	0.95	1.02	0.99	1.02	0.99
Move Scale						
Intracounty	0.94	0.90	0.99	0.93	0.98	0.93
Intrastate	**1.23**	**0.97**	**1.06**	**1.06**	**1.07**	**1.05**
Interstate	**1.07**	**0.97**	**1.06**	**1.13**	**1.06**	**1.09**

Sources: National Change of Address Registry (2019-2021) linked to IRS 1040s (Tax Year 2020) and Census Numident (2022). CBDRB-FY2022-CES005-018.
Notes: Address changes for 2020 and 2021 are normalized by 2019 address changes and should be interpreted as ratios. Values greater than 1 indicate an increase relative to 2019, while values less than 1 indicate a decrease relative to 2019. To highlight meaningful patterns, increases in 2020 and 2021 greater than the national average by more than 1 percentage point (+0.01) are shown in boldface font.

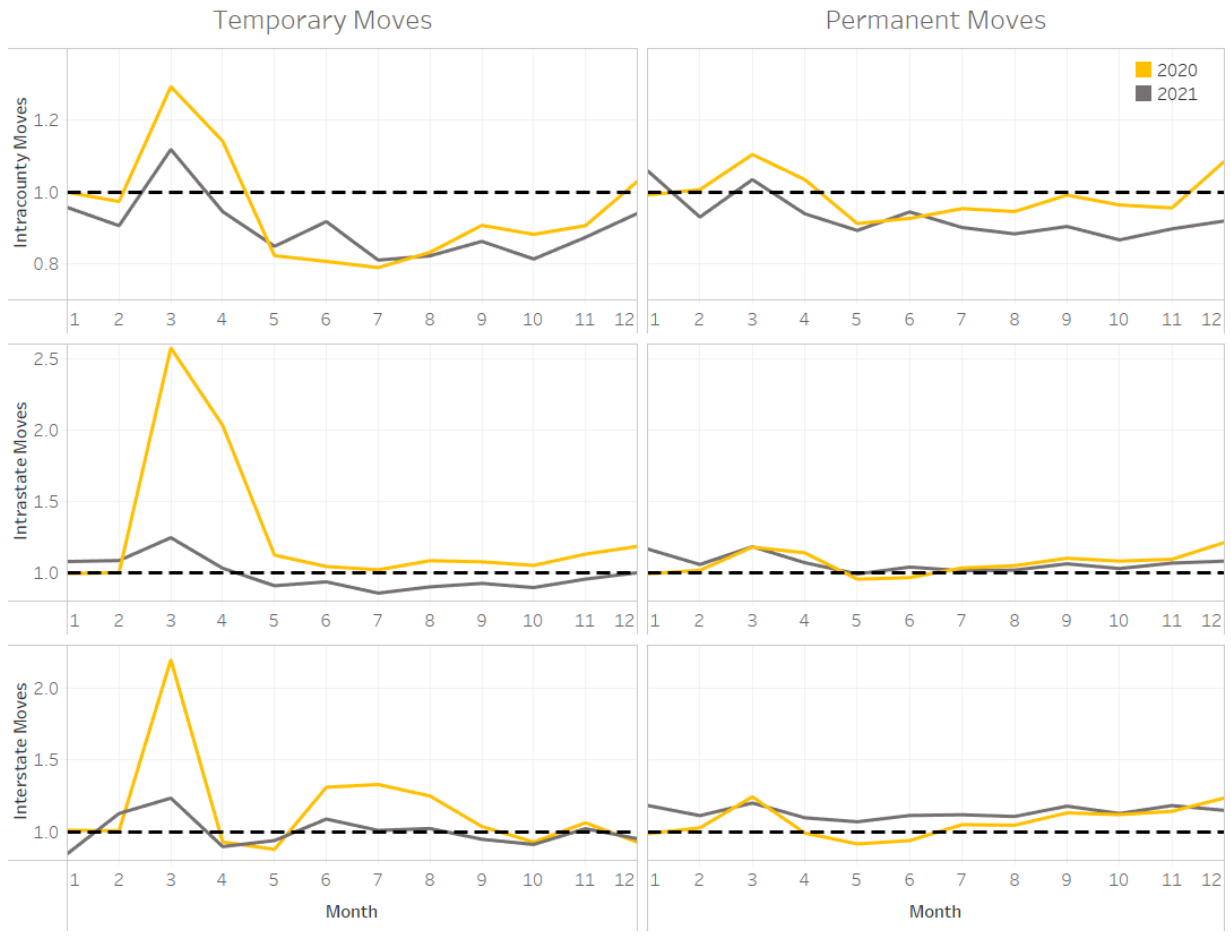
The Geography of Address Changes, 2019-2021

The pandemic migration spike was driven by a sustained increase in moves across state and county boundaries rather than local moves within counties. Table 3 disaggregates 2020 and 2021 change of address counts by a rough measure of move scale. Consistent with ACS trends, intracounty address changes – moves that do not cross a county border – were 2% and 7% lower in 2020 and 2021 than in 2019, but we find a shift toward intrastate (i.e., moves across county borders within the same state) and interstate moves not captured by the ACS that is sustained through 2021. Relative to 2019, intrastate address changes increased 7% in 2020 and 5% in 2021, while interstate address changes increased by 6% in 2020 and 9% in 2021. It is important to note, however, that these shifts are relative to 2019 and do not, for example, suggest that intrastate or interstate moves were more prevalent than local, intracounty moves during the pandemic. Despite a decline in prevalence relative to 2019, intracounty moves remain more common than moves across county and state lines through 2020 and 2021. (Please refer to Appendix Table 1 for information on the baseline 2019 distribution of address changes by move scale and other characteristics discussed below.)

This shift in geographic scale is common to both temporary and permanent address changes, but monthly trends reveal that the shift is driven primarily by permanent movers. Figure 1 reports monthly address changes relative to 2019 by scale and duration. We see large increases in both temporary and permanent migration across all move scales in March of 2020, the month a national state of emergency was declared: temporary address changes increased relative to 2019 by 29%, 57%, and 20% at the intracounty, intrastate, and interstate scales, respectively, while permanent address changes increased by 11%, 18%, and 24%, respectively. Temporary and permanent migration at the local level fell below 2019 levels thereafter, but remained slightly elevated at the intra- and interstate levels through most of 2020. Sustained increases in larger-scale movement through 2021, however, are seen only among permanent migrants moving across state lines.

The simultaneous increase in both temporary and permanent moves early in the pandemic tenuously suggests that these two types of address changes are largely treated as fundamentally distinct by movers. In the aggregate, we cannot conclude that those who would have otherwise moved permanently moved temporarily instead because of the pandemic. That said, we also cannot rule out the possibility that such behavior was employed among some segments of the population. Temporary migration increased more between 2019 and 2020 than permanent migration (6% vs. 2%), suggesting that some new pool of temporary movers was produced in response to the pandemic. Some of these movers may have been permanent movers in the hypothetical absence of the pandemic, but others are new movers spurred only by the novel pandemic conditions. We return to this question below in our discussion of migrant selection by AGI rank.

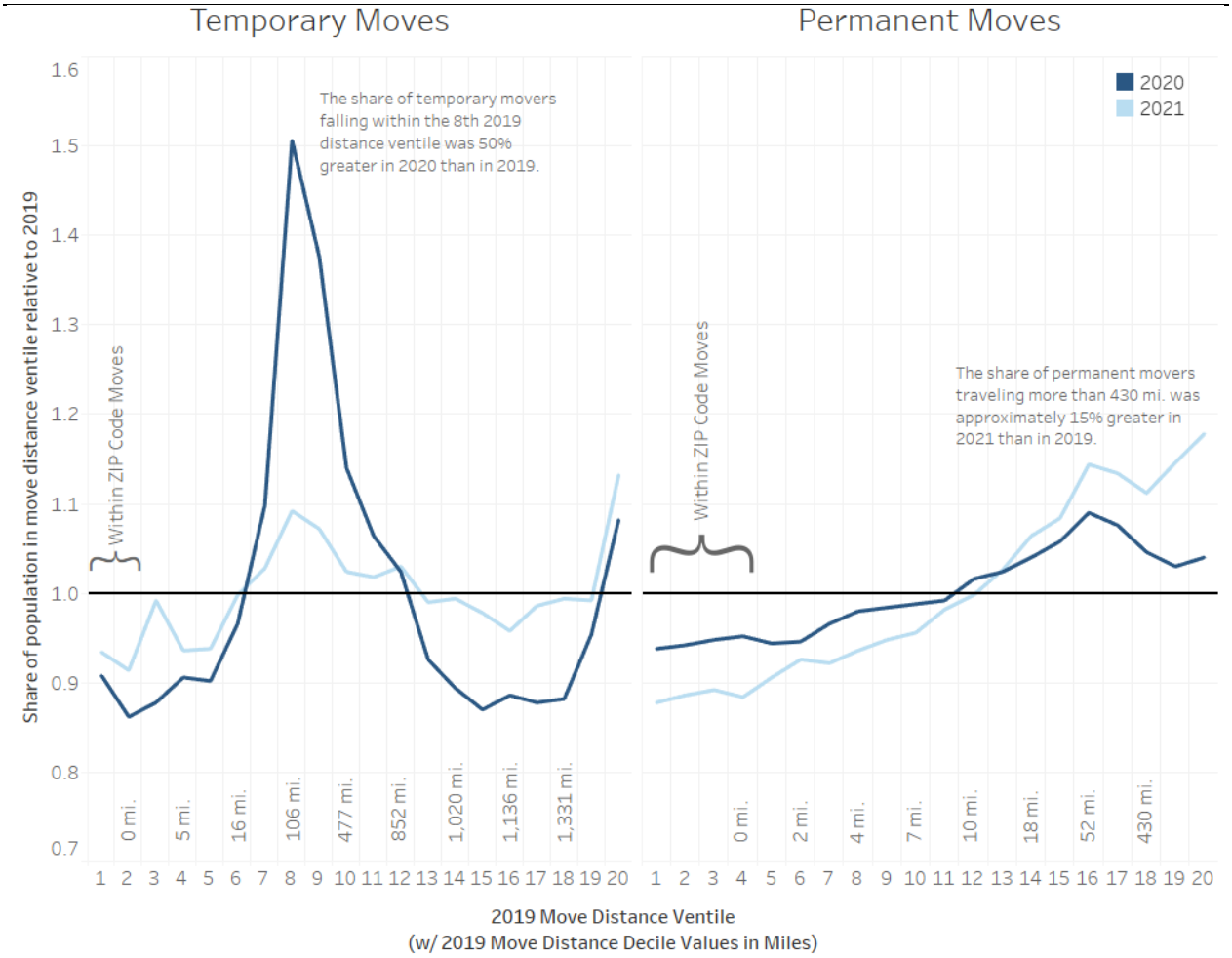
Figure 1: Temporary and Permanent Address Changes by Scale and Month, 2020 and 2021 relative to 2019



Source: National Change of Address Registry (2019-2021). CBDRB-FY2022-CES005-018.
Notes: Intracounty, intrastate, and interstate move categories are mutually-exclusive and are assigned using a restricted 9-digit zip code-to-county FIPS conversion file developed by the U.S. Census Bureau. Dashed reference lines at y=1 represent 2019 values to which 2020 and 2021 values are compared.

Regardless of the precise relationship between temporary and permanent moves, the shift in favor of larger-scales is not merely an artifact of haphazard county and state boundaries – that is, it is not driven by increased migration across lines that pass through large metropolitan areas or commuting zones. Rather, Americans changing addresses in 2020 and 2021 moved farther than those moving in 2019. Figure 2 compares the distances between origin and destination ZIP code centroids for those changing addresses in 2019, 2020, and 2021 and is constructed as follows. First, we take a random 5% sample of all address changes in 2019 and rank them into move distance ventiles; by definition, 5% of the 2019 sample falls into each ventile bin. Second, we apply the 2019 ventile distance breaks to random 5% samples of all moves from 2020 and 2021, and we note the share of 2020 and 2021 address changes falling into each 2019 ventile bin. Finally, we calculate the ratio of 2020 and 2021 population shares in each ventile to the share in 2019 (5%) and plot those ratios. Values above 1.0 indicate that the population whose move

Figure 2: Distribution of Move Distances, 2020 and 2021 relative to 2019



Source: Five-percent random sample of NCOA moves (2019-2021). CBDRB-FY24-CES010-010.
Notes: These graphs are constructed as follows. First, 2019 moves are ranked into ventiles by distance between origin and destination ZIP code centroids; moves within a ZIP code are assigned a move distance of 0 miles. Second, distances at each ventile break are noted and decile move distances in miles are included along the x-axis. Note that these distance breaks are calculated separately for temporary and permanent address changes. Third, we calculate the share of 2020 and 2021 moves falling within each 2019 move distance ventile. Finally, we normalize the 2020 and 2021 population shares in each 2019 distance bin by the 2019 population share in that bin (by definition, 5%) and plot these ratios in darker and lighter blue, respectively. The reference line at y = 1 indicates the 2019 population shares. Plotted values above 1.0 indicate that the population whose move distance fell in a given 2019 distance ventile increased in 2020 or 2021, while values below 1.0 indicate a decline relative to 2019.

distance fell in a given 2019 distance bin increased, while values below 1.0 indicate a decline relative to 2019.

Increases in the distance traveled by NCOA movers are primarily a function of a steady shift toward longer distance permanent, rather than temporary, moves in 2020 and through 2021. As shown in the right-hand panel of Figure 2, the share of all permanent moves falling in the bottom 20% of the 2019 distance distribution (which were entirely within-ZIP-code moves covering a distance of 0 miles), fell by 6.2% in 2020 and by 12.2% in 2021. At the same time, the share of

moves in the top 10% of the 2019 distance distribution (moves covering 430 miles or more) increased by 4% in 2020 and by 17.8% in 2021. Changes in temporary move distances (left-hand panel of Figure 2) during the pandemic are less clear and consistent. We find a massive 50.6% increase in temporary moves in the 8th ventile (40-44.9th percentiles) of the 2019 distance distribution (moves of just over 100 miles) in 2020 and a sustained increase of 8% and 13.2% in the top 5% of the 2019 move distance distribution (moves covering more than 1,300 miles) in 2020 and 2021. Short-distance temporary moves in the bottom third of the 2019 distance distribution (moves under approximately 16 miles), as well as moves between approximately 800 and 1,200 miles, fell in 2020 but had returned to near 2019 levels by 2021. Broadly speaking, these findings are consistent with those from Sullivan and Genadek (2024: 21-22) who, using a composite administrative record file created by the Census Bureau, document a 9.5 mile (roughly 6%) increase in mean move distances between 2019 and 2021.

Moreover, the shift toward larger-scale, longer-distance moves is accompanied by a geographic shift in the flow of migrants across broad Census divisions. Tables 4 and 5 report changes in the size of interregional flows of temporary and permanent migrants, respectively. (Baseline regional move distributions for 2019 are available in Appendix Table 2.) Consistent with media narratives, we find evidence of a 34% increase in temporary migration from the Middle Atlantic states (including New York) in 2020 toward New England and North Central states (Table 4). These shifts in temporary flows were moderated but otherwise sustained into 2021 and accompanied by an increase in temporary flows away from Pacific states to a variety of regions, but particularly to Southern states. In many ways, the geographic shifts in temporary migration mirror migration flows witnessed after other natural disasters: migrants appear to be fleeing (or, as shown on the diagonal, increasingly moving within) regions with large population centers which were hit hard by the pandemic early on and whose densities may have been ideal for COVID-19's spread. Temporary migration within most Census divisions declined to lower than 2019 levels in 2021, but the continued disruption of temporary migration patterns across divisions into 2021 is surprising considering that excess mortality had largely curbed by March of 2021 (Foster et al. 2024).[14]

The shift in temporary migration from Middle Atlantic and Pacific states toward New England and the South is mirrored among permanent migrants (Table 5). Again, consistent with media narratives, we find evidence of a 14% increase in the total outflow of migrants from the Middle Atlantic region and a 39% increase in permanent migrants from the Middle Atlantic to New England, specifically. We see a similar 15% increase in permanent outflows from the Pacific region, but Pacific migrants chose a variety of destinations across the U.S. with a slight preference for the South that continued through 2021. Interestingly, though temporary migration out the Middle Atlantic and Pacific Regions was accompanied by increased movement within those regions as well, we find that permanent moves within these regions decreased relative to 2019 in both 2020 and 2021.

[14] See also NCHS estimates of excess mortality at https://www.cdc.gov/nchs/nvss/vsrr/covid19/excess_deaths.htm.

Table 4: Temporary Interregional Migration Flows, 2020 and 2021 relative to 2019

	2020 Destination									
2020 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	1.16	1.19	1.45	1.39	0.93	1.14	1.10	0.99	1.01	**0.98**
Middle Atlantic	3.57	1.31	2.81	2.88	1.11	1.88	1.74	1.32	1.69	**1.34**
East North Central	1.28	1.30	1.00	1.30	0.95	0.99	0.94	0.95	0.99	**0.97**
West North Central	1.11	1.09	1.29	1.01	0.97	0.90	0.86	0.93	0.98	**0.96**
South Atlantic	0.93	0.91	0.93	1.02	1.03	1.11	1.16	1.07	1.01	**0.95**
East South Central	1.08	1.10	1.06	1.04	1.00	1.05	1.14	1.02	0.94	**1.02**
West South Central	1.15	1.13	1.07	1.05	1.17	1.24	1.07	1.10	1.11	**1.11**
Mountain	0.91	0.89	0.92	0.94	1.02	1.14	1.04	0.99	0.95	**0.96**
Pacific	1.49	1.37	1.44	1.27	1.42	1.48	1.46	1.14	1.10	**1.29**
Total Inflow	**1.21**	**0.97**	**1.04**	**1.09**	**1.02**	**1.14**	**1.09**	**1.04**	**1.05**	

	2021 Destination									
2021 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	0.93	1.01	1.11	1.09	0.90	1.03	1.01	0.92	0.99	**0.93**
Middle Atlantic	1.37	0.94	1.22	1.32	0.95	1.15	1.16	0.99	1.09	**0.99**
East North Central	0.93	1.00	0.87	0.97	0.91	0.84	0.88	0.88	0.89	**0.90**
West North Central	1.15	1.06	1.00	0.88	0.99	0.84	0.86	0.92	0.92	**0.94**
South Atlantic	0.95	0.93	0.94	1.02	0.94	0.97	1.07	1.02	1.00	**0.95**
East South Central	1.01	0.96	0.95	0.96	0.91	0.94	0.97	0.98	0.95	**0.94**
West South Central	1.15	1.16	1.00	0.96	1.11	1.15	1.02	1.02	1.08	**1.06**
Mountain	1.00	1.00	0.94	0.94	1.05	1.10	1.00	0.93	0.93	**0.96**
Pacific	1.37	1.33	1.26	1.08	1.43	1.36	1.40	1.02	0.96	**1.19**
Total Inflow	**1.01**	**0.96**	**0.97**	**0.99**	**0.95**	**0.98**	**1.00**	**0.96**	**0.96**	

Source: National Change of Address Registry (2019-2021). CBDRB-FY24-CES010-010.

Notes: Counts of 2020 temporary migrants between origin and destination Census divisions normalized by 2019 flows are shown. Values above 1.0 indicate an increase in migrant flows relative to 2019 (in blue, with larger increases in darker shades), while values below 1.0 indicate a decrease in migrant flows relative to 2019 (in red, with larger declines in darker shades). Cells on the diagonal capture migrant flows within and between states and counties in each division.

Table 5: Permanent Interregional Migration Flows, 2020 and 2021 relative to 2019

	2020 Destination									
2020 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	0.97	1.02	1.05	1.06	1.00	1.07	1.03	1.03	0.94	**1.01**
Middle Atlantic	1.39	0.98	1.14	1.16	1.12	1.13	1.13	1.13	1.05	**1.14**
East North Central	0.98	0.95	0.97	1.01	1.02	1.03	1.03	1.00	0.90	**1.00**
West North Central	0.98	0.93	1.05	0.99	1.01	1.02	1.02	0.98	0.92	**1.00**
South Atlantic	1.04	1.00	1.02	1.05	1.03	1.06	1.04	1.02	0.94	**1.02**
East South Central	1.01	0.98	1.02	1.08	1.02	1.02	1.06	1.05	0.96	**1.02**
West South Central	1.02	0.95	1.04	1.06	1.03	1.07	1.05	1.05	0.96	**1.03**
Mountain	1.07	0.99	1.09	1.09	1.08	1.14	1.10	1.04	1.01	**1.06**
Pacific	1.14	1.03	1.14	1.16	1.16	1.20	1.19	1.16	1.02	**1.15**
Total Inflow	**1.14**	**0.99**	**1.06**	**1.07**	**1.07**	**1.08**	**1.08**	**1.08**	**0.97**	

	2021 Destination									
2021 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	0.94	1.14	1.10	1.12	1.18	1.16	1.24	1.09	1.03	**1.14**
Middle Atlantic	1.26	0.95	1.10	1.06	1.27	1.20	1.26	1.11	0.99	**1.20**
East North Central	1.09	1.06	0.94	1.02	1.16	1.09	1.18	1.05	1.01	**1.10**
West North Central	1.16	1.06	1.08	0.95	1.16	1.10	1.10	1.01	0.99	**1.08**
South Atlantic	1.01	1.02	1.01	1.05	0.99	1.11	1.12	1.07	0.98	**1.04**
East South Central	1.11	1.05	1.05	1.07	1.10	0.97	1.11	1.08	0.99	**1.08**
West South Central	1.15	1.12	1.09	1.08	1.15	1.14	1.01	1.07	1.03	**1.10**
Mountain	1.23	1.17	1.18	1.17	1.34	1.39	1.29	0.98	1.08	**1.20**
Pacific	1.21	1.19	1.20	1.25	1.40	1.57	1.52	1.22	0.98	**1.31**
Total Inflow	**1.13**	**1.07**	**1.08**	**1.10**	**1.23**	**1.17**	**1.25**	**1.12**	**1.02**	

Source: National Change of Address Registry (2019-2021). CBDRB-FY24-CES010-010.

Notes: Counts of 2020 permanent migrants between origin and destination Census divisions normalized by 2019 flows are shown. Values above 1.0 indicate an increase in migrant flows relative to 2019 (in blue, with larger increases in darker shades), while values below 1.0 indicate a decrease in migrant flows relative to 2019 (in red, with larger declines in darker shades). Cells on the diagonal capture migrant flows within and between states and counties in each division.

To some degree, the fact that geographic shifts among permanent movers mirror those among temporary migrants is not surprising. As shown in Table 1, we find a steady increase in the share of all moves made by those who moved at some point earlier in the year; some of those repeat movers may have first moved only temporarily when the pandemic hit, but subsequently made their address change permanent or otherwise moved again permanently. We return to this point below in our discussion of income selection among migrants. But the rise in repeat moves is not sufficient to fully explain the geographic shifts we see and, thus, the new trends we report challenge both the environmental migration literature and Willekens and Baydar's (1983) assertion that geographic patterns in migration are remarkably stable once levels are controlled.

The Composition of the Population Changing Addresses, 2019-2021

The literature on migration during environmental and economic upheavals suggests that the COVID-19 pandemic will produce increases in migration among demographic and socioeconomic groups on the margins and/or those potentially most vulnerable to COVID-19. As shown in Table 6 – which reports changes in temporary and permanent migration by age, race, ethnicity, move type, and within-birth-cohort Adjusted Gross Income (AGI) in tax year 2020 – we find, in some cases, exactly the opposite.

Consistent with the vulnerability hypothesis, we find that individual (as opposed to family) moves, moves among younger Americans (those under 25 in particular),[15] and moves among some racial and ethnic minority groups[16] increased during the pandemic. These groups may have found themselves vulnerable to the economic downturn as the pandemic set in (Polyakova et al. 2020) – working in industries and occupations less amenable to telework and/or deemed non-essential – or subject to the shuttering of colleges and universities. Moreover, movers changing addresses on their own may have been more vulnerable than families leaning on each other for support. Broadly, these group-specific findings are consistent with prior research utilizing CPS-ASEC data and noting an increase in young adults moving in with their parents.[17] Our findings are unique, though, in that they highlight a sustained increase in migration not among non-Hispanic White Americans (Quealy 2020) but, instead, among ethnoracial minorities, in general, and those self-identifying as Hispanic, non-Hispanic American Indian or Alaska Native, non-

[15] We report monthly trends in 2020 change of address increases by age group in Appendix Figure 1, which shows that the annual trends reported in Table 5 are quite stable across the entire year.
[16] We report monthly trends in 2020 change of address increases by race and ethnicity in Appendix Figure 2, which reveals some within-year variation in temporary migration by race/ethnicity that is not borne out in Table 5. Specifically, Appendix Figure 2 shows that the March 2020 spike in temporary migration was largest in the NH Multiple Race, NH Asian alone, and (despite their below-national-average ratio in Table 5) NH White alone populations at approximately 110% greater than 2019, but lowest in the NH Black alone population at 35%. Broadly speaking, these findings are somewhat consistent with the notion that temporary migrants early in the pandemic were positively selected with respect to sociodemographic characteristics.
[17] For example, see Cohen's analysis of IPUMS-CPS data: https://familyinequality.wordpress.com/2020/07/06/why-there-are-2-9-million-extra-young-adults-living-at-home/ and https://familyinequality.word-press.com/2020/08/16/july-data-show-2-7-million-extra-young-adults-living-at-home/.

Table 6: Select Characteristics of NCOA Address Change Filers and Their Moves, 2020 and 2021 relative to 2019

	Temporary		Permanent		Overall	
	2020	2021	2020	2021	2020	2021
Total	1.06	0.95	1.02	0.99	1.02	0.99
Move Type						
Individual	**1.16**	**1.01**	**1.04**	**1.02**	**1.05**	**1.02**
Family	0.94	0.88	0.95	0.93	0.95	0.92
Mover's Age Group						
Under 25	**1.44**	**1.41**	**1.36**	**1.65**	**1.36**	**1.64**
25 to 44	**1.28**	**1.04**	**1.04**	0.98	**1.05**	0.99
45 to 64	**1.25**	**1.16**	0.98	0.95	1.00	0.97
65 to 84	0.93	0.89	0.98	0.97	0.96	0.94
85+	0.89	0.75	0.92	0.88	0.91	0.86
Mover's Race and Ethnicity						
Missing	**1.17**	**1.25**	**1.10**	**1.22**	**1.11**	**1.22**
Hispanic	**1.12**	**1.04**	**1.05**	**1.03**	**1.06**	**1.03**
NH White Alone	1.05	0.92	0.99	0.96	1.00	0.95
NH Black Alone	1.02	**0.99**	1.03	1.01	1.03	**1.01**
NH AIAN Alone	1.07	**1.03**	**1.04**	**1.03**	**1.04**	**1.03**
NH Asian Alone	**1.21**	**1.03**	0.99	**1.01**	1.01	**1.01**
NH NHPI Alone	**1.08**	**1.07**	**1.07**	**1.04**	**1.07**	**1.04**
NH SOR Alone	1.02	0.90	1.02	**1.01**	1.02	1.00
NH Multiple Race	**1.21**	**1.03**	**1.05**	**1.05**	**1.06**	**1.05**
Mover's Age-Adjusted AGI Rank (TY 2020)						
Non-Filers	0.96	0.85	0.95	0.91	0.95	0.90
Bottom Decile	**1.10**	**1.00**	**1.07**	**1.07**	**1.08**	**1.06**
Second Decile	1.05	0.95	**1.06**	**1.04**	**1.06**	**1.03**
Third Decile	1.01	0.93	1.02	**1.01**	1.02	1.00
Fourth Decile	1.01	0.93	1.01	1.00	1.01	1.00
Fifth Decile	1.01	0.93	1.00	1.00	1.00	1.00
Sixth Decile	1.02	0.94	1.01	**1.01**	1.01	1.00
Seventh Decile	1.03	0.95	1.01	**1.02**	1.01	**1.01**
Eighth Decile	1.04	0.95	1.02	**1.02**	1.02	**1.01**
Ninth Decile	1.08	**0.97**	**1.04**	**1.01**	**1.04**	**1.01**
Top Decile	**1.28**	**1.09**	**1.09**	1.00	**1.12**	**1.02**

Sources: National Change of Address Registry (2019-2021) linked to IRS 1040s (Tax Year 2020) and Census Numident (2022). CBDRB-FY22-CES005-018, CBDRB-FY24-CES010-010, and CBDRB-FY24-CES010-015.
Notes: Adjusted Gross Incomes for tax year 2020 are measured at the household (tax unit) level and, contingent on filing, ranked within birth year cohorts. Address changes for 2020 and 2021 are normalized by 2019 address changes and should be interpreted as ratios. Values greater than 1 indicate an increase relative to 2019, while values less than 1 indicate a decrease relative to 2019. To highlight meaningful patterns, increases in 2020 and 2021 greater than the national average by more than 1 percentage point (+0.01) are shown in boldface font.

Hispanic Native Hawaiian or Pacific Islander, and non-Hispanic Multiple Race, in particular (see also Cohn 2020, 2021).

However, results by tax year 2020 AGI rank within birth-year cohorts suggest that migration during the pandemic was, to a large extent, *bifurcated* with respect to economic vulnerability, increasing most among those at the extremes of the income distribution. The age-adjusted income ranks reported in Table 6 facilitate comparisons of migration behavior by household or tax unit income for individuals of the same age; for example, they rank the incomes of college students against all other college students, retirees against all other retirees, and so on. Note that one's income rank is determined not just among other movers, but among all peers born in the same year regardless of migration status.

Increases in temporary address changes in 2020 were common to all those who filed taxes, but they were largest among those in the bottom (10% increase) and top (28% increase) deciles. Likewise, while no filers saw a decline in permanent migration in 2020, increases were most pronounced among those in the top and bottom quintiles. Through 2021, temporary moves remained elevated relative to 2019 for only those in the top income decile (9%), while permanent moves remained most elevated among those in the bottom quintile of filers. As such, we find some support for the notion that migration during the pandemic was a function of economic vulnerability and security, and that both vulnerability and security seem to have had a positive effect on migration. By a large margin, however, both temporary and permanent migration declined the most among those who did not file tax year 2020 returns: overall, we find a 5% drop in 2020 and a 10% drop in 2021 among non-filers relative to 2019. To the degree that non-filing reflects economic vulnerability (i.e., income below the filing requirement threshold) or is associated with other demographic factors that might inhibit migration (e.g., old age), then, the story becomes more complicated.

Monthly trends in temporary and permanent address changes shown in Figure 3 shed more light on the relationship between economic vulnerability and migration. Throughout 2020, the positive income gradient on temporary migration is clear: the left-hand panel of Figure 3 shows that temporary address changes increased monotonically as tax year 2020 AGI increased, and temporary migration relative to 2019 was almost always lowest among non-filers. Broadly speaking, those with the means to do so moved temporarily both as the pandemic began in early 2020 and as it wore on later in the year. Note, however, that temporary migration increased by 40% in March of 2020 even among non-filers. The relationship between AGI rank and permanent migration, however, changed dramatically over the course of 2020. In the first half of the year, there was an inverse relationship between permanent address changes and migration: permanent moves increased most among those in the bottom half of the AGI distribution and decreased by 15% among top earners in May 2020. In the second half of 2020, however, this relationship reversed such that permanent moves among top earners increased by about 25% relative to 2019, but moves among non-filers and those in the bottom 40% of earners declined relative to 2019.

Figure 3: Temporary and Permanent Address Changes by Adjusted Gross Income Rank and Month, 2020 relative to 2019



Source: National Change of Address Registry (2019-2020) linked with IRS 1040s (tax year 2020) and Census Numident (2022). CBDRB-FY42-CES010-015.
Notes: Adjusted Gross Incomes for tax year 2020 are measured at the household (tax unit) level and, contingent on filing, ranked within birth year cohorts. They are, therefore, adjusted for expected increases in income over the life course. Dashed reference lines at y=1 represent 2019 values to which 2020 values are compared. Note that y-axis scales differ across panels.

It is difficult to disentangle income-driven patterns in temporary and permanent migration shown in Figure 3 from any potential simultaneous shifts in the relative prevalence of temporary and permanent moves. But the monthly trends by AGI rank provide some evidence that temporary address changes may have served as a precursor to or substitute for permanent migration for some early in the pandemic. Among those in the top income decile rank for their age, fewer permanent moves were made in the first 6 months of 2020 than in the first six months of 2019. But, over this same period, temporary migration among those in the top income decile for their age increased dramatically. One plausible interpretation of these patterns is that relatively high earners used temporary migration as a substitute for more permanent relocation, particularly in the first half of 2020 when uncertainty was high, but translated these temporary moves into

permanent relocations as the pandemic wore on, as employers' acceptance of full-time telework spread, and as mortgage interest rates continued to fall through January of 2021. Such patterns are not apparent among those in lower AGI ranks: these populations saw a relatively small increase in temporary migration contained entirely to March of 2020, but a large increase in permanent migration in March and April of 2020, followed by a decline in permanent migration relative to 2019 through November 2020. In short, the interchangeability of temporary and permanent moves may be a function of economic resources: while lower earners appear to have been forced into permanent and lasting migration decisions as the pandemic began, higher earners delayed these decisions until later in 2020 – first moving temporarily, then translating these temporary moves into more permanent relocations when advantageous.

Taken together, subgroup-specific patterns suggest some relationship between migration and vulnerability during the pandemic, but they also indicate subtle shifts in this relationship over time. Early in 2020, although migration increased among nearly all sociodemographic groups (with the exception of family movers and those 65 and over), increased migration was most concentrated among the relatively advantaged (high earners, non-Hispanic White populations) and those least vulnerable to complications from COVID-19 (younger Americans). Table 6 (and Appendix Figure 1) suggest that younger Americans may have been particularly impacted in March 2020 by college closures, but the sustained increase in migration among those under 25 throughout 2021 is more difficult to explain. Moving into 2021, we find a sustained increase in permanent address changes among racial and ethnic minority groups and those at the bottom of the income distribution (non-filers notwithstanding), groups which may have been more vulnerable to the longer-lasting economic impacts of the pandemic. Finally, monthly results by income rank in Figure 3 point to the flexibility relatively high earners enjoyed when navigating migration choices during the pandemic. As such, though we find evidence consistent with prior work on migration in response to economic and environmental disasters, this vulnerability hypothesis only partially fits migration patterns in the novel and complicated pandemic context.

Understanding Discrepancies between Administrative and Census Survey Migration Trends

We shift focus now to the apparent failure of the ACS and CPS-ASEC to capture aggregate 2020 increases in permanent changes of address, which likely stems from three sources acting alone or in concert. First, discrepancies may indicate that survey and NCOA migration measures differ conceptually. Survey migration measures may overlook repeat and/or temporary moves captured in the NCOA registry. However, results reported in Table 2 suggest that even without a 2020 increase in temporary address changes and the repeat mover share, more permanent changes of address were filed in 2020 than in 2019. In other words, conceptual differences in the measurement of migration appear to be insufficient to explain the aggregate discrepancies we report here, but they could contribute to them nonetheless.

Table 7: ACS and CPS-ASEC Misreporting of Permanent Interstate Address Changes, 2020 and 2021

	What share of all householders filing a permanent change of address report that move in their survey response?	
	2020	**2021**
ACS	67.2%	69.7%
CPS-ASEC	49.4%	49.5%

Source: National Change of Address registry (2019-2021) linked with 1-year ACS (2020-2021) and CPS-ASEC (2020-2021) responses. CBDRB-FY2022-CES005-018 and CBDRB-FY24-CES010-001.
Notes: The linked samples included in this table are restricted to ACS and CPS-ASEC householders with unedited (non-allocated, non-imputed) survey responses whose 1-year migration reference period covers a permanent change of address found in the NCOA registry. Survey respondents filing only a temporary change of address are excluded, as temporary relocations are less likely to correspond conceptually with a change in place of usual residence captured in the ACS and CPS-ASEC.

Second, survey-NCOA discrepancies may be a symptom of declining survey response rates, in general, and a severe spike in nonresponse during the pandemic, in particular. Between 2005 and 2022, ACS response rates declined from 97.3% to 84.4%, while CPS-ASEC response rates declined even more precipitously from 82.6% to 61.4% (Foster, Ellis, and Fiorio 2023). Over the same period, migration item non-response also increased such that, as of 2021, only 75.3% and 48.7% of all those sampled for the ACS and CPS-ASEC, respectively, answered the "residence one year ago" question (ibid.). Research assessing survey nonresponse suggests that recent movers are less likely to respond to surveys (Dillon 2021; Foster 2020), so rising non-response over time means that surveys will increasingly underestimate migration. Moreover, pandemic-specific assessments find that nonresponse increased disproportionately among younger populations (Rothbaum et al. 2021) – the population among which changes of address increased the most in 2020 (Table 6). As such, the divergence in 2020 between survey and NCOA migration trends may indicate that NCOA captures moves among population subgroups less likely to respond to the ACS and CPS-ASEC. Estimating the contribution of nonresponse to discrepancies is beyond our scope here, but we acknowledge that this is an important contributor to the gap.

A third possibility, to which we now turn, is that permanent changes of address filed with the USPS are not accurately reported by individual ACS and CPS-ASEC survey respondents. Table 7 restricts analysis to ACS and CPS-ASEC householders who also filed a permanent change of address with the USPS in the one-year period referenced in their unedited survey migration response. While we would expect a very high rate of agreement among this restricted sample, we find that only around two-thirds of ACS householders and fewer than half of all CPS-ASEC householders requesting a permanent address change in the year leading up to their survey response reported living at a different residence one year ago on their survey. Among CPS-ASEC householders permanently changing addresses and recording that change with the USPS, the share also reporting their move on the survey remained stable at 49.4% and 49.5% in 2020 and 2021, respectively. Among analogous ACS householders, however, we find some evidence that the misreporting of permanent address changes may have increased in 2020: the share

correctly reporting their permanent address change increased from 67.2% in 2020 to 69.7% in 2021.

Broadly, these findings are consistent with prior work documenting the misreporting of program participation, income sources, and income amounts on Census Bureau surveys (Bee and Mitchell 2017; Meyer and Mittag 2015; Rothbaum 2015), as well as migration-specific work comparing CPS-ASEC migration responses with migration measures constructed from Longitudinal Employer-Household Dynamics data (Hyatt et al. 2018) and ACS and CPS-ASEC migration responses with migration measures constructed from IRS 1040s and information returns (Foster et al. 2018b, 2023). Taken together with this prior work, our results suggest that the apparent misreporting of migration on Census Bureau surveys is not a pandemic-specific phenomena, but a trend that began in the early 1990s and has grown since.

Discussion

Our analysis demonstrates the utility of administrative data as both a check on and complement to Census Bureau survey measures, particularly as the costs and difficulties of addressing survey and item nonresponse increase over time. Though administrative data have been used to assess the accuracy of a broad range of survey responses, their relatively untapped potential for measuring migration and residential mobility have been singled out as particularly promising (Keller et al. 2022). As we show here, administrative data can provide insights into temporary as well as permanent migration at monthly (or even weekly or daily) intervals in a relatively timely fashion. That said, administrative data pose their own unique challenges and limitations. Broadly speaking, individual-level linkages of the sort we perform here depend upon the assignment of linkage keys (PIKs). USPS data, specifically, carry additional challenges stemming from the fact that migrants are not required to file a change of address. We know of no prior work on the factors governing selection into the NCOA data and, as explained above, were unable to offer such an assessment here. As a result, we operate throughout with the assumption that selection pressures were unchanged during the pandemic.

We find that internal migration, as measured by voluntary change of address requests filed with the USPS, increased by 2% during the COVID-19 pandemic, particularly in March and April of 2020 among younger, higher-income individuals making longer-distance moves. Although this aggregate increase was not sustained through 2021, changes in the composition and geography of migrant streams it introduced were. The pandemic had a unique, profound, and, at least through its second year, persistent effect on the spatiality of American migration and residential mobility. The endurance of these spatial impacts in the post-pandemic years will be an important topic to investigate with future releases of administrative data.

These patterns reflect the reactions of Americans to the novel and quickly evolving questions posed by the global pandemic, and they differ in important ways from migration responses to natural disasters and economic recessions in the recent past. Broadly, increased migration in the aggregate during the pandemic economic downturn – and increased migration among the highest

as well as the lowest earners – largely reverses trends witnessed during the Great Recession. Like migration from disaster-affected areas, migration during the pandemic was characterized by increased flows out of regions with large, dense urban areas. Unlike disaster migration, however, we find little evidence (at least through 2021) that return migration reversed these trends; instead, 2021 patterns suggest continued flows out of the Mid-Atlantic and Pacific toward the South and other traditionally less common destinations across the U.S. Pandemic migration also differed from natural disaster migration with respect to the composition of migrant flows. While disaster flight typically increases as a function of vulnerability and means, pandemic patterns saw *reduced* migration among those economically (e.g., non-filers) and physically (e.g., 65+ years of age) vulnerable and increased migration among the young and those at the top and bottom of the income distribution.

The migration-relevant mechanisms acting during the pandemic clearly differed from those operating in other natural and economic disasters, but it remains unclear precisely what drove pandemic era migration changes. Directions for future research are numerous. Increased migration among young individuals in higher AGI tax units at the start of the pandemic point to campus closures as an important factor. Increased tolerance for telework in some occupations likely contributed to the spike in permanent migration among higher earners in the second half of 2020, but the shift to longer distance migration throughout 2021 may also reflect these workers' "Great Resignations" when called back to the office (Cohen 2021). Economic Impact Payments from the CARES Act, Consolidated Appropriations Act, and American Rescue Plan Act[18] and federal moratoria on evictions may have eased migration pressures among non-filers and lower income Americans, affording them housing stability in the midst of an otherwise unstable economy.

Though understanding these mechanisms is important for migration theory and population health, we focused in this paper on a more fundamental task: establishing whether and how migration changed during the pandemic. This seemingly simple task is complicated by the fact that well-established migration measures from the ACS and CPS-ASEC suggesting a decline in migration contradict both our collective pandemic experiences (reflected in headlines like those quoted at the start) and a swath of less traditional administrative data suggesting increased mobility. By linking ACS and CPS-ASEC data to change of address filings at the individual level, we show that at least part of this discrepancy is rooted in the apparent failure of survey householders to accurately report permanent address changes they filed with USPS during their 1-year survey reference period. We are not the first to document this tendency to misreport migration, and we stress that it is not a pandemic-specific phenomenon likely to correct itself in the near future: prior work establishes that CPS-ASEC migration underreporting dates to at least the early 1990s and appears to be increasing over time (Hyatt et al. 2018). Migration misreporting appears to be less pronounced in the ACS than in the CPS-ASEC but is also increasing over time (Foster et al. 2023). When coupled with rising nonresponse rates, the ability of these surveys to accurately capture migration trends, particularly in response to acute shocks,

[18] https://www.pandemicoversight.gov/data-interactive-tools/data-stories/update-three-rounds-stimulus-checks-see-how-many-went-out-and

may be compromised. We will continue this ongoing line of research examining migration misreporting in the ACS, CPS-ASEC, and other household surveys since 2000 using IRS 1040s, IRS information returns, and other administrative data.

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Appendix Table 1: The Distribution of NCOA Moves by Various Individual, Household, and Move Characteristics, 2019

	Temporary	Permanent	Overall
Mover's Race/Ethnicity			
Missing	2.70	5.93	5.58
Hispanic	6.32	12.77	12.08
NH White	79.13	61.78	63.65
NH Black	7.53	13.22	12.61
NH AIAN	0.52	0.84	0.80
NH Asian	2.44	3.28	3.19
NH NHPI	0.09	0.17	0.16
NH SOR	0.35	0.26	0.27
NH Multiple	0.91	1.76	1.67
Mover's Age Group			
Under 25	2.27	5.25	4.93
25 to 44	18.54	47.82	44.67
45 to 64	15.51	21.66	21.00
65 to 84	44.76	8.90	12.77
85+	18.91	16.37	16.64
Mover's Age-Adjusted AGI Rank (TY 2020)			
Non-Filers	18.74	22.06	21.74
Bottom Decile	6.63	7.18	7.13
Second Decile	6.79	8.26	8.11
Third Decile	6.45	8.26	8.09
Fourth Decile	6.52	8.14	7.98
Fifth Decile	7.48	8.73	8.61
Sixth Decile	6.53	6.87	6.84
Seventh Decile	8.59	8.09	8.14
Eighth Decile	7.91	6.48	6.62
Ninth Decile	11.10	8.06	8.35
Top Decile	13.26	7.88	8.40
Type of Move			
Individual	51.62	70.62	68.57
Family	48.38	29.38	31.43
Scale of Move			
Intracounty	23.40	60.22	56.25
Intrastate	13.15	22.12	21.15
Interstate	63.44	17.67	22.60

Source: National Change of Address Registry (2019) linked to IRS 1040s (Tax Year 2020) and Census Numident (2022). CBDRB-FY2024-CES010-015.

Notes: Adjusted Gross Incomes for tax year 2020 are measured at the household (tax unit) level and, contingent on filing, ranked within mover birth year cohorts.

Appendix Table 2: Distribution of NCOA Moves by Origin and Destination Census Division, 2019

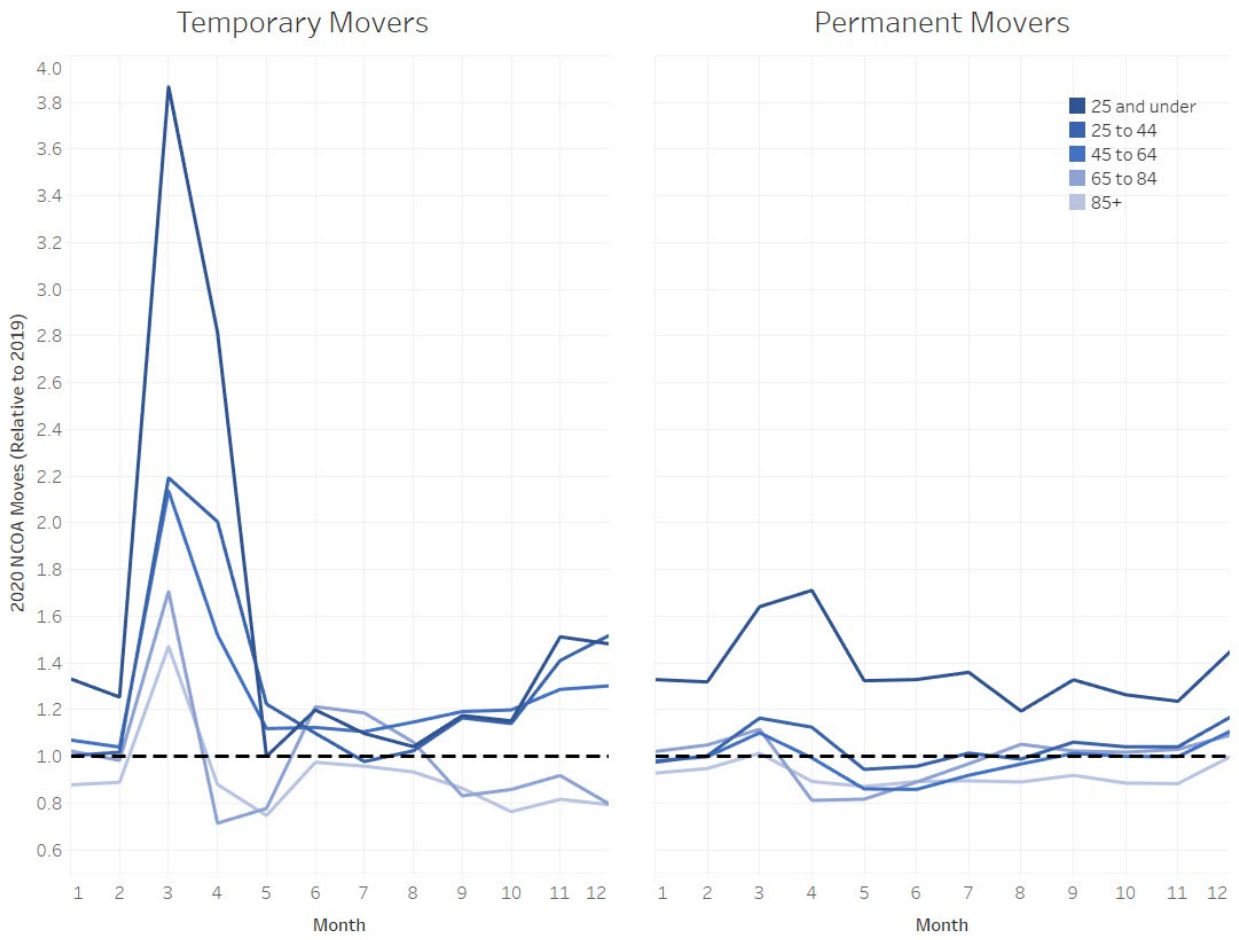
Temporary Moves	2019 Destination									
2019 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	5.08	0.59	0.15	0.06	6.66	0.07	0.18	0.42	0.41	**8.54**
Middle Atlantic	0.67	11.32	0.29	0.09	11.34	0.14	0.31	0.68	0.66	**14.19**
East North Central	0.21	0.32	14.67	0.42	13.29	0.64	1.06	2.86	0.92	**19.72**
West North Central	0.07	0.10	0.40	6.08	2.53	0.20	1.18	3.28	0.61	**8.36**
South Atlantic	5.78	8.56	7.88	1.58	25.23	1.01	0.66	1.11	1.05	**27.63**
East South Central	0.08	0.13	0.37	0.10	1.39	3.70	0.20	0.17	0.16	**2.59**
West South Central	0.17	0.23	0.55	0.49	0.64	0.16	8.14	0.68	0.45	**3.36**
Mountain	0.39	0.56	1.76	1.69	1.10	0.15	0.67	10.89	2.93	**9.24**
Pacific	0.33	0.46	0.59	0.38	0.95	0.14	0.47	3.04	14.90	**6.37**
Total Inflow	**7.69**	**10.95**	**11.99**	**4.81**	**37.91**	**2.51**	**4.73**	**12.23**	**7.19**	100.00

Permanent Moves	2019 Destination									
2019 Origin	New England	Middle Atlantic	East North Central	West North Central	South Atlantic	East South Central	West South Central	Mountain	Pacific	**Total Outflow**
New England	4.54	1.04	0.36	0.14	2.42	0.16	0.32	0.38	0.63	**5.45**
Middle Atlantic	1.33	10.78	1.03	0.31	6.48	0.43	0.84	0.81	1.45	**12.68**
East North Central	0.35	0.88	15.21	1.51	4.37	1.48	1.37	1.67	1.47	**13.10**
West North Central	0.14	0.28	1.48	7.34	1.37	0.37	1.19	1.34	0.84	**7.00**
South Atlantic	1.90	4.57	4.23	1.17	21.34	2.59	2.35	1.87	2.41	**21.07**
East South Central	0.12	0.29	1.26	0.32	2.42	5.52	0.90	0.41	0.43	**6.15**
West South Central	0.24	0.56	1.06	1.05	2.19	0.92	11.44	1.52	1.43	**8.97**
Mountain	0.32	0.59	1.37	1.36	1.82	0.45	1.63	8.00	3.38	**10.93**
Pacific	0.57	1.12	1.36	0.88	2.73	0.64	2.18	5.16	15.82	**14.65**
Total Inflow	**4.97**	**9.31**	**12.14**	**6.75**	**23.80**	**7.04**	**10.78**	**13.17**	**12.05**	100.00

Source: National Change of Address Registry (2019). CBDRB-FY24-CES010-015.

Notes: Cells on the diagonal capture migrant flows within and between states and counties in each division.

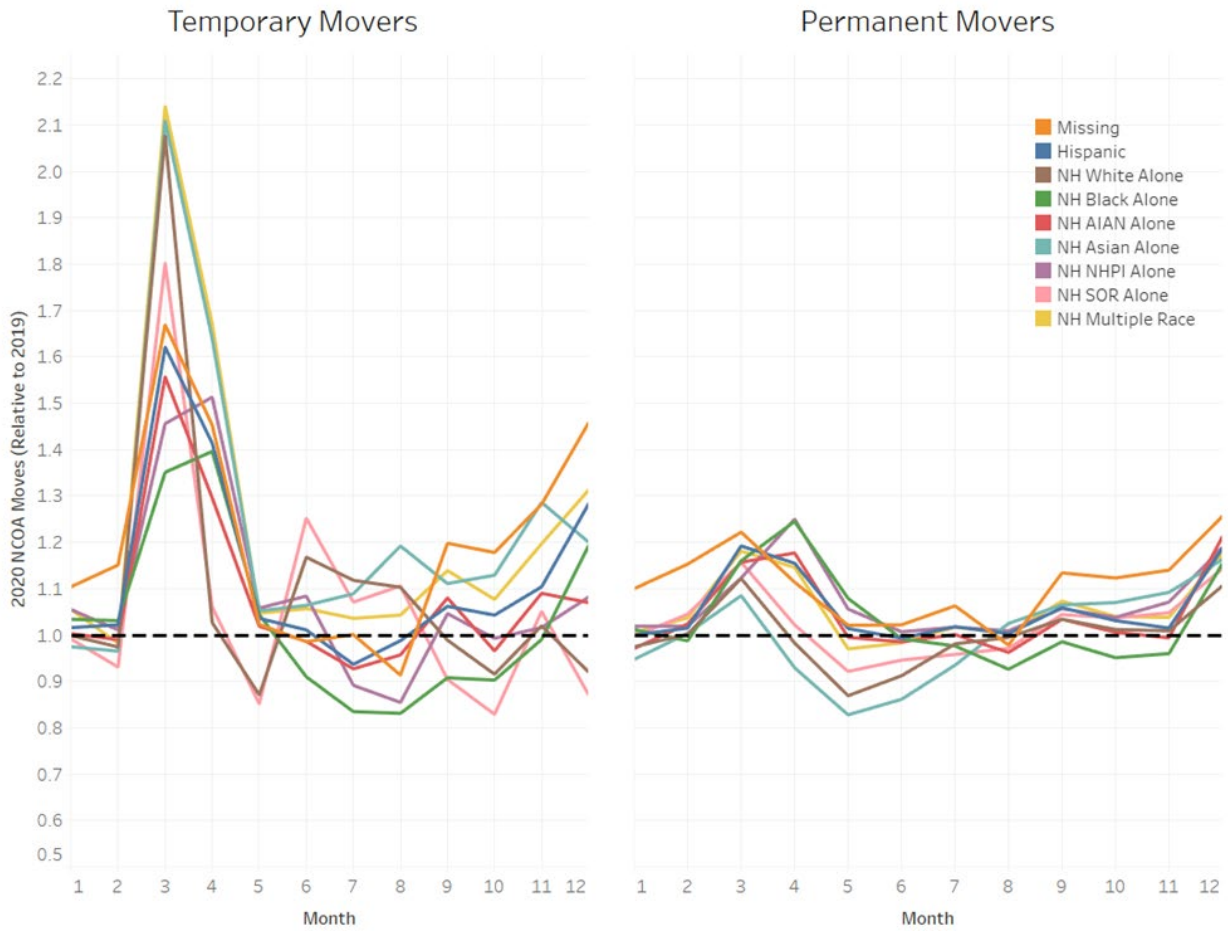
Appendix Figure 1: Temporary and Permanent Address Changes by Age Group and Month, 2020 relative to 2019



Source: National Change of Address Registry (2019-2020) linked with Census Numident (2022). CBDRB-FY2022-CES005-018.

Notes: Dashed reference lines at y=1 represent 2019 values to which 2020 values are compared.

Appendix Figure 2: Temporary and Permanent Address Changes by Race, Ethnicity, and Month, 2020 relative to 2019



Source: National Change of Address Registry (2019-2020) linked with the Census Best Race File (2020). CBDRB-FY24-CES010-010.
Notes: Dashed reference lines at y=1 represent 2019 values to which 2020 values are compared.